Abdelrahman Abdelwahab

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EXPERIENCE

Software Engineer III May 2026 - Present
TomTom *Amsterdam*

- Restored HD-map release gate reporting by rewiring a Databricks/PySpark validation job to per-tile logs, enabling full metrics across four road-class tiers.
- Removed the human approval step from failed-work recovery (design) by designing a Delta Lake failure ledger, transient/persistent classifier, and auto-routed Jira tickets, replacing a linearly growing approval queue with a fixed ceiling of two batch slots per day.
- Built twice-daily map-publishing volume dashboard automation with KQL / Azure Log Analytics, Python, and GitHub Actions, replacing manual tracking and restoring throughput visibility.

Software Development Engineer Feb 2025 - Mar 2026
Amazon *London*

- Owned end-to-end development and production launch of simulation tooling used to model operational strategies, delivering a full-stack solution (React, Java, AWS ECS, Lambda) that reduced planning time by 70%.
- Engineered the multi-region launch path for Japan by deploying new AWS CDK infrastructure and managing critical Liquibase migrations for data platform and backend services.
- Implemented progressive delivery with feature flags (AWS SSM Parameter Store) and end-to-end CloudWatch metrics and alarms, enabling safer rollouts and faster incident triage.

Software Development Engineer Intern Apr 2024 - Oct 2024
Amazon *Luxembourg*

- Automated release approval workflows using AWS Step Functions, SNS, SES, Lambda, and SQS, eliminating manual handoffs and saving ~1 day per release on average.
- Built a React, Java, PostgreSQL rule management system with bulk upload and audit tracking, improving operational turnaround time by ~15%.

Software Engineering Internship Dec 2022 - Jun 2023
Sysdig Inc. *Remote, Italy*

- Designed and executed an automated testing pipeline with infrastructure provisioning (Terraform, AWS, Ansible), plugin testing, reporting, and teardown, accelerating release cycles by 20%.
- Created an integration tool using Xray and Jira API, connecting Sysdig test repositories to improve workflow tracking, reporting, and collaboration; adopted by multiple teams.
- Extended the testing pipeline with an automated reporting stage, giving teams detailed test environment and outcome reports for faster release decision-making.

EDUCATION

Politecnico di Torino Sep 2021 - Dec 2025
BSc. in Computer Engineering *Turin, Italy*

PROJECTS

World Cup Ghorba | React, TanStack Start, TypeScript, Supabase, Vercel, Bun, Three.js

- Built a full-stack World Cup 2026 prediction game for a private friend-group league, including locked predictions, leaderboard scoring, admin result submission, player auth, profiles, and match dashboards.
- Implemented server workflows for live result sync, SMS prediction reminders, head-to-head duels, reactions, qualifier bonuses, and live match questions using TanStack server functions, Supabase, and Vercel.

Video Sharing Platform | Next.js, TypeScript, Express.js, GCP, Firebase, Pub/Sub, FFmpeg

- Built a cloud-native video pipeline on GCP using Cloud Storage, Pub/Sub, Cloud Run, Cloud Functions, Firestore, and FFmpeg for asynchronous upload, transcoding, hosting, and metadata management.
- Created a responsive Next.js web app hosted on Cloud Run, integrating Firebase Functions APIs for video retrieval and playback workflows.

Security System | OpenCV, Flask, S3, Twilio, React

- Developed a Python security system using OpenCV for person detection, integrating S3 for remote video storage and Twilio for real-time notifications.
- Designed a React control interface with arm/disarm flows and activity logs, improving operational visibility and monitoring of system state.

TECHNICAL SKILLS

Languages: Java, Python, C/C++, JavaScript/TypeScript, SQL, KQL
Backend & Web: React, Next.js, Express.js, Django, Flask, REST APIs, Node.js
Cloud & Infrastructure: AWS (ECS, Lambda, CDK, S3, SSM, Step Functions, SQS/SNS/SES, CloudWatch), Azure (Databricks, Delta Lake, Log Analytics), GCP (Cloud Run, Pub/Sub, Cloud Functions, Firestore, Cloud Storage), Docker, Kubernetes, Terraform, Ansible, Liquibase
Reliability & Operations: Observability, metrics/alerts, release automation, progressive delivery, feature flags, PySpark, Linux, CI/CD, GitHub Actions, incident triage
Databases: PostgreSQL, Firebase (Firestore)
Tools: Git, Jira, Xray

ACHIEVEMENTS

Finalist: The 2021 ICPC Southwestern Europe Regional Contest - SWERC
Qualified: The 2021 Italian Collegiate Programming Contest - ITACPC